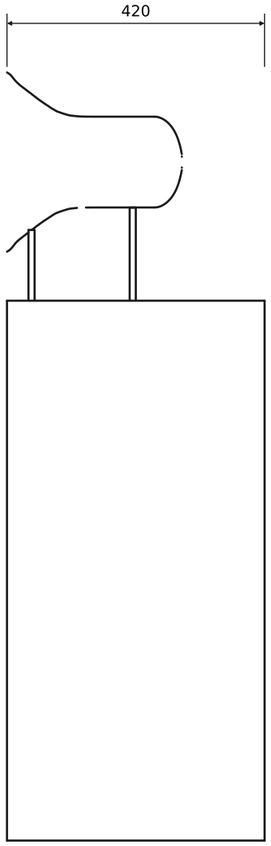
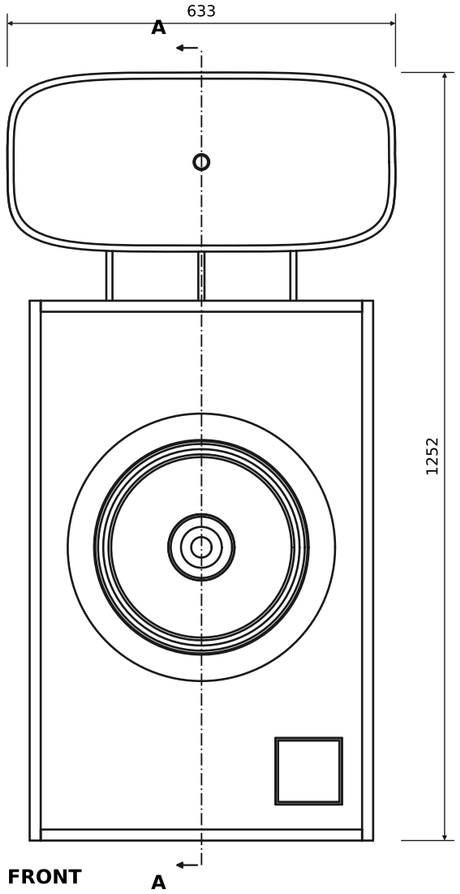
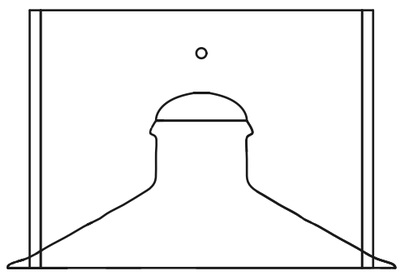
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

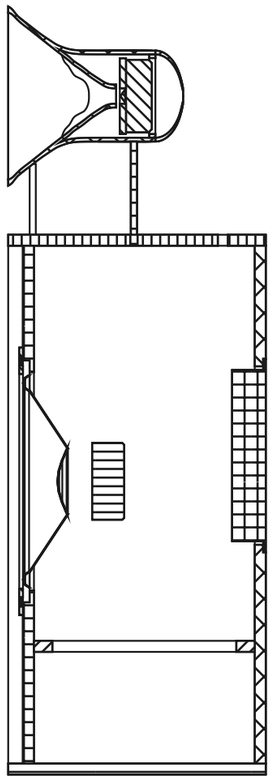
General arrangement

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.2 and seen from the right; cut faces hatched, a different angle for each part.



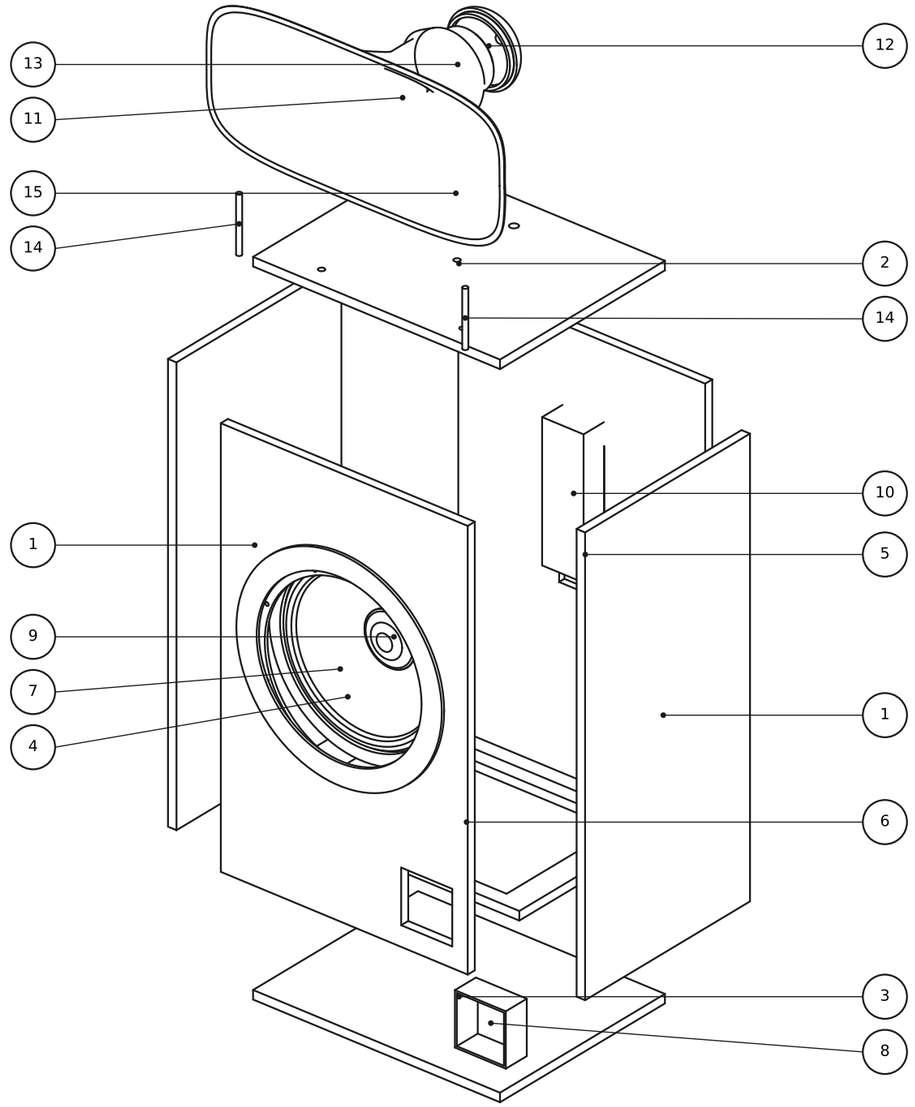
Section A-A

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



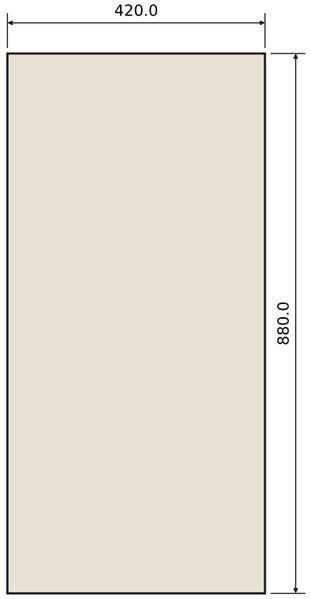
Exploded view

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

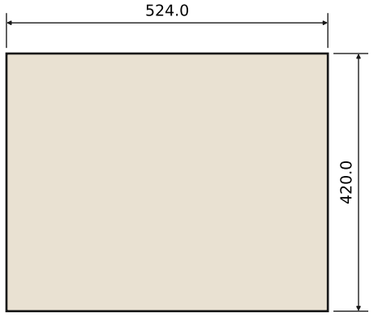
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



left, right

18 mm baltic-birch, 4 for the set



bottom

18 mm baltic-birch, 2 for the set

Sheet parts: plain panels

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

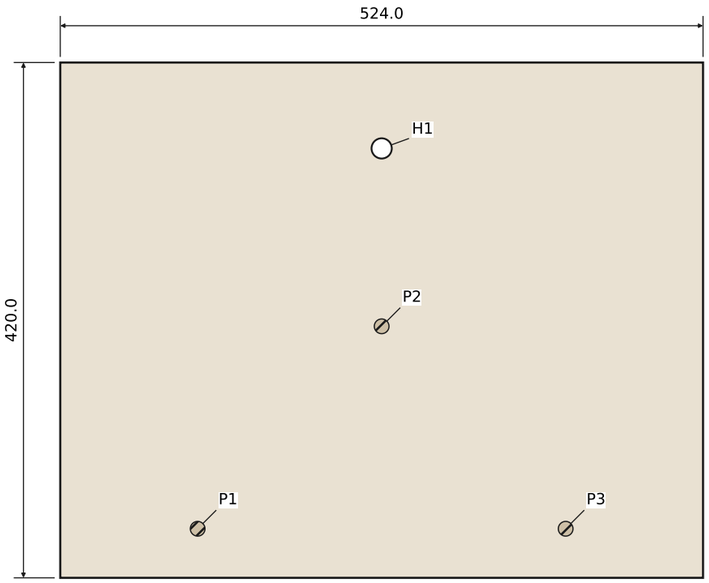
mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/top.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø12.0	112.0	40.0	15 from face
P2	Ø12.0	262.0	205.0	15 from face
P3	Ø12.0	412.0	40.0	15 from face
H1	Ø16.5	262.0	350.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



top: 18 mm baltic-birch, 2 for the set, seen from its outer face

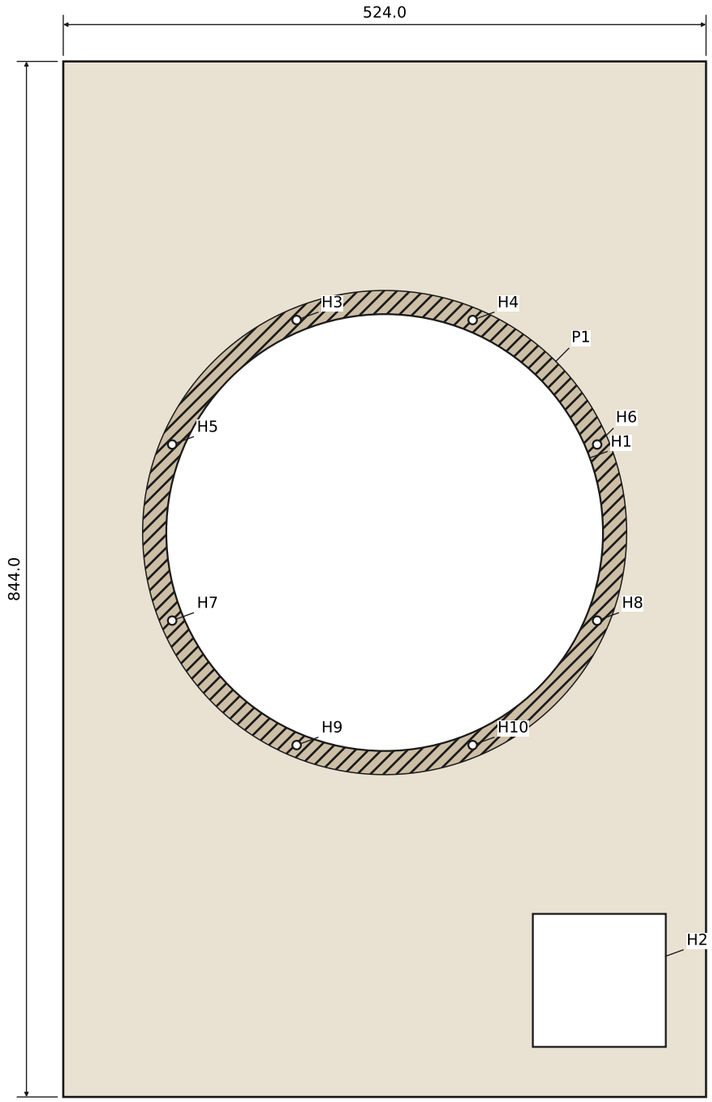
Sheet part: top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/front-panel.dxf. Marks listed at the right.



front-panel: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø394.6	262.0	460.0	11 from face
H1	Ø356.0	262.0	460.0	through
H2	108.4 x 108.4	437.0	95.0	through
H3	Ø7.0	190.2	633.2	through
H4	Ø7.0	333.8	633.2	through
H5	Ø7.0	88.8	531.8	through
H6	Ø7.0	435.2	531.8	through
H7	Ø7.0	88.8	388.2	through
H8	Ø7.0	435.2	388.2	through
H9	Ø7.0	190.2	286.8	through
H10	Ø7.0	333.8	286.8	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

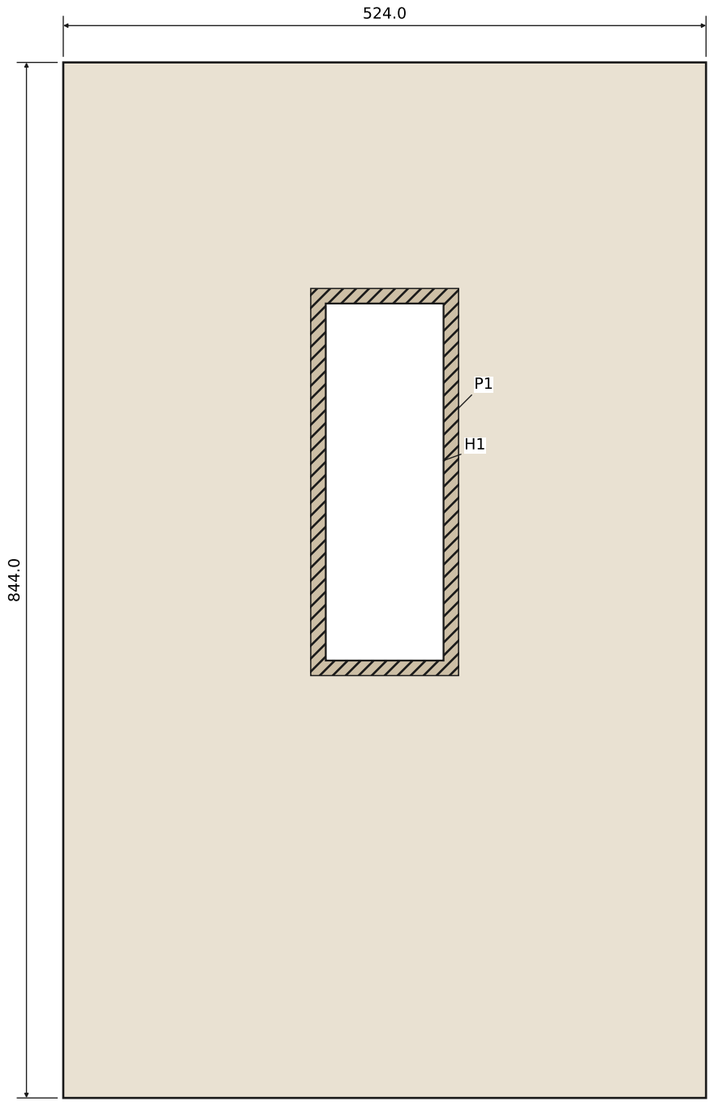
Sheet part: front-panel

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.



back: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 x 315.5	262.0	502.0	4.5 from face
H1	96.0 x 291.0	262.0	502.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

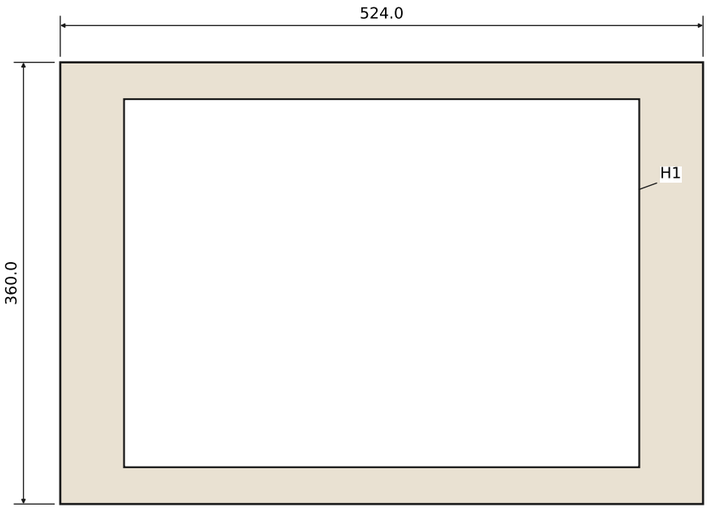
mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 300.0	262.0	180.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace: 18 mm baltic-birch, 2 for the set, seen from its outer face

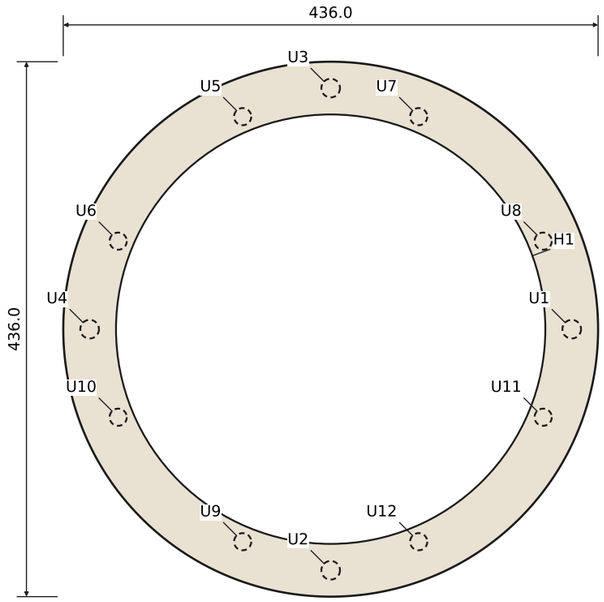
Sheet part: brace

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/woofer-ring.dxf. Marks listed at the right.



woofer-ring: 6 mm mdf, 2 for the set, seen from its outer face

inner-face pockets: dxf/woofer-ring-underside.dxf (seen from that face)

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	414.5	218.0	3.2 from inside
U2	Ø15.2	218.0	21.5	3.2 from inside
U3	Ø15.2	218.0	414.5	3.2 from inside
U4	Ø15.2	21.5	218.0	3.2 from inside
U5	Ø14.0	146.3	391.2	4 from inside
U6	Ø14.0	44.8	289.8	4 from inside
U7	Ø14.0	289.8	391.2	4 from inside
U8	Ø14.0	391.2	289.8	4 from inside
U9	Ø14.0	146.3	44.8	4 from inside
U10	Ø14.0	44.8	146.2	4 from inside
U11	Ø14.0	391.2	146.2	4 from inside
U12	Ø14.0	289.8	44.8	4 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

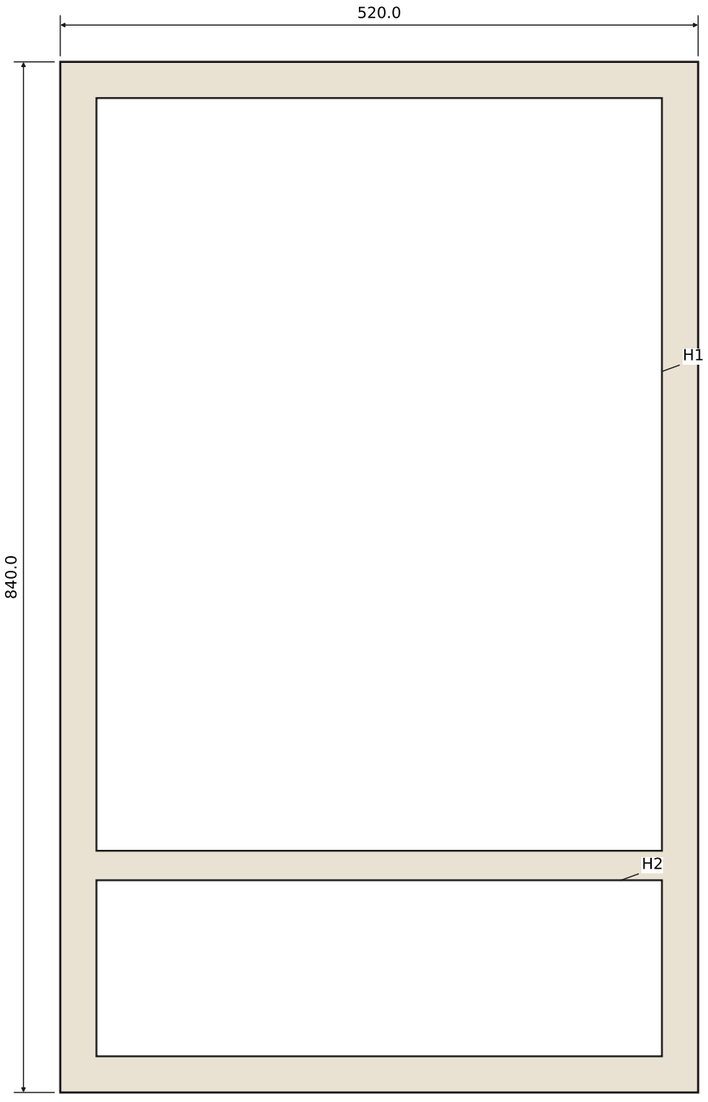
Sheet part: woofer-ring

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/cloth-frame.dxf. Marks listed at the right.



cloth-frame: 6 mm mdf, 2 for the set, seen from its outer face (option: cloth)

mark	size	x	y	depth
H1	461.0 x 613.5	260.0	503.8	through
H2	461.0 x 143.5	260.0	101.2	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

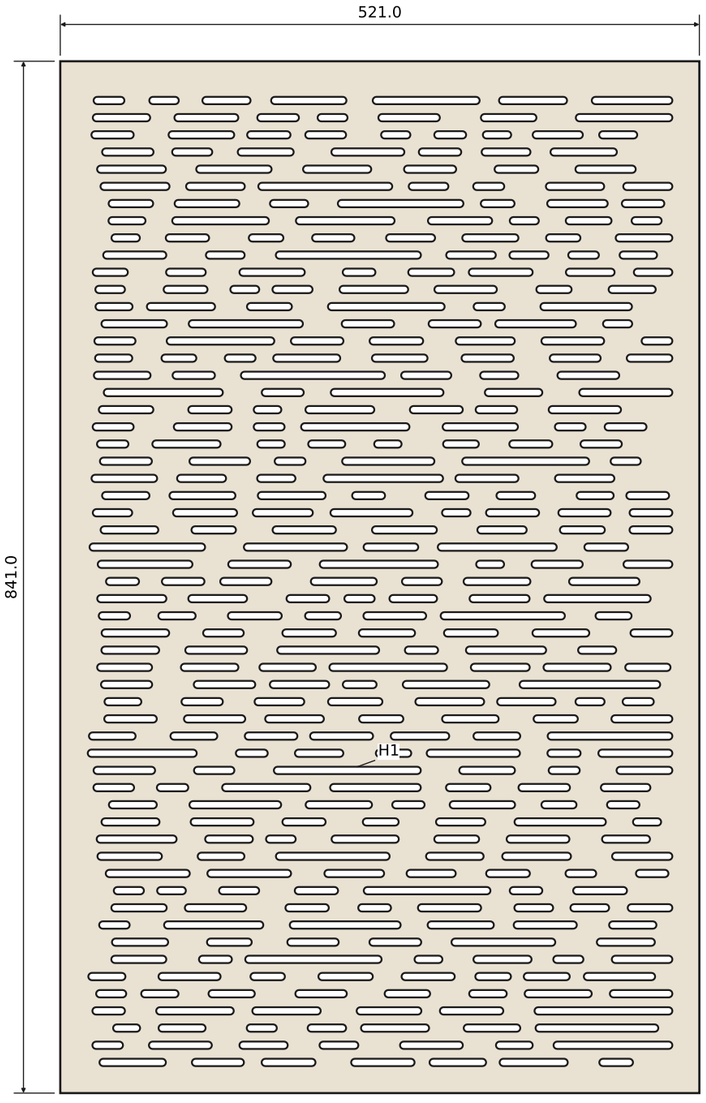
Sheet part: cloth-frame (option: cloth)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/grille-lenticel.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	slot 6.0 wide, 22-120 long	394 off	DXF	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

grille-lenticel: 6 mm baltic-birch, 2 for the set, seen from its outer face (option: lenticel-grille)

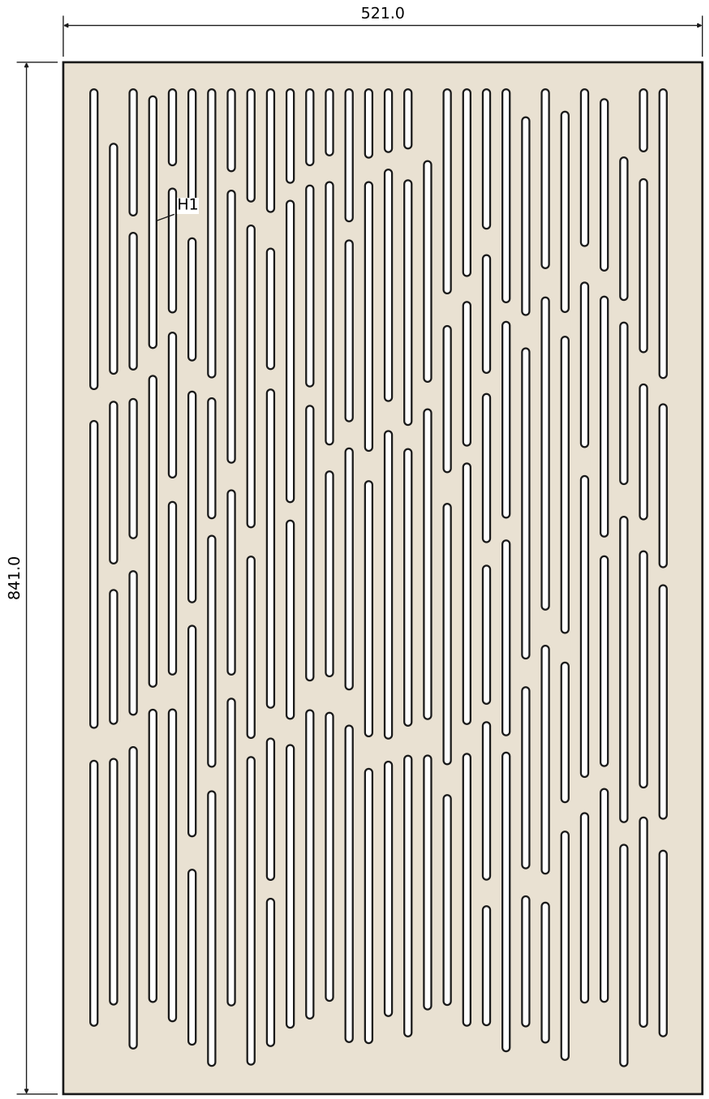
Sheet part: grille-lenticel (option: lenticel-grille)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/grille-strips.dxf. Marks listed at the right.



grille-strips: 6 mm spotted-gum-ply, 2 for the set, seen from its outer face (option: strip-grille)

mark	size	x	y	depth
H1	slot 6.0 wide, 48-259 long	124 off	DXF	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: grille-strips (option: strip-grille)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

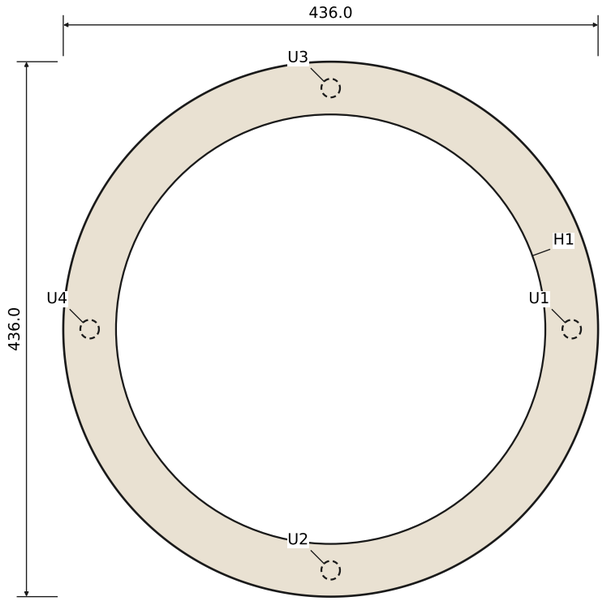
mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/disc-frame.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	414.5	218.0	3.2 from inside
U2	Ø15.2	218.0	21.5	3.2 from inside
U3	Ø15.2	218.0	414.5	3.2 from inside
U4	Ø15.2	21.5	218.0	3.2 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



disc-frame: 12 mm mdf, 2 for the set, seen from its outer face (option: cloth-disc)
inner-face pockets: dxf/disc-frame-underside.dxf (seen from that face)

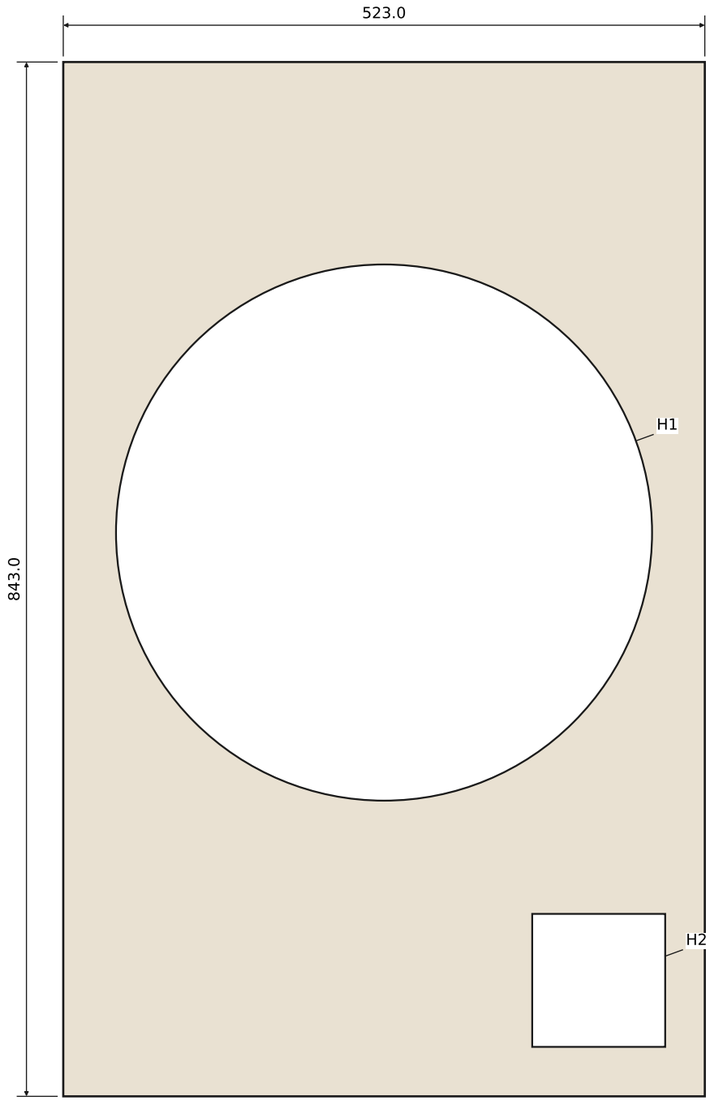
Sheet part: disc-frame (option: cloth-disc)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/cork-face.dxf. Marks listed at the right.



cork-face: 3 mm cork, 2 for the set, seen from its outer face (option: cork-face)

mark	size	x	y	depth
H1	Ø437.0	261.5	459.5	through
H2	108.4 x 108.4	436.5	94.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: cork-face (option: cork-face)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

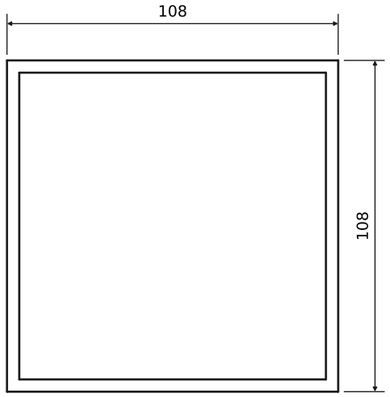
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

BAR-014 rev A

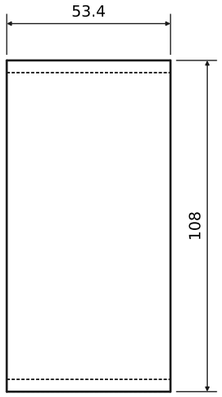
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

8. port: asa print, on end; 89 cm3 each; 2 for the set; 1:2

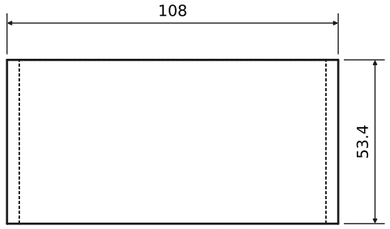
100 square, 53 long from the panel's face: tunes the box to 38 Hz; print 15 longer and trim to tune



FRONT



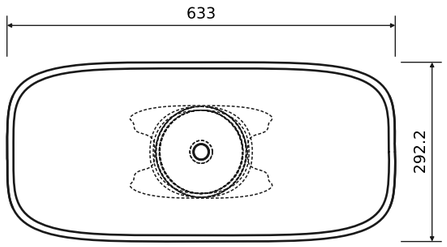
FROM THE RIGHT



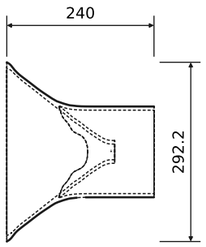
FROM ABOVE

11. waveguide: cast, cast-resin; 1513 cm3 each; 2 for the set; 1:10

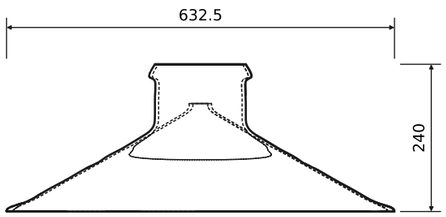
mouth 612 x 272, 170 deep, 1 in throat; a shell 6 thick following its flare, 10 at the rim, and a pod 148 across behind it round the driver; cast in tinted translucent polyurethane in a two-part silicone mould from a CNC-cut plug, bead-blasted to frost; M8 inserts bonded into pads inside its underside take the rods



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

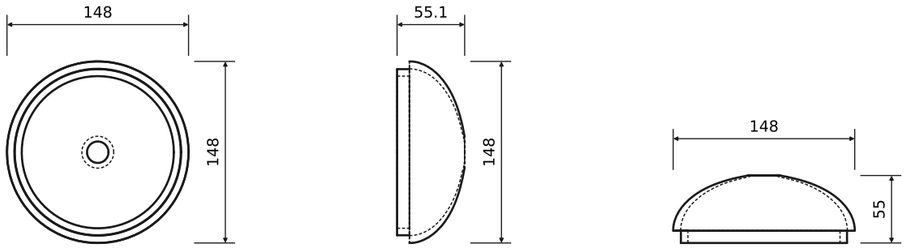
mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

12. waveguide-cap: cast, cast-resin; 117 cm3 each; 2 for the set; 1:5

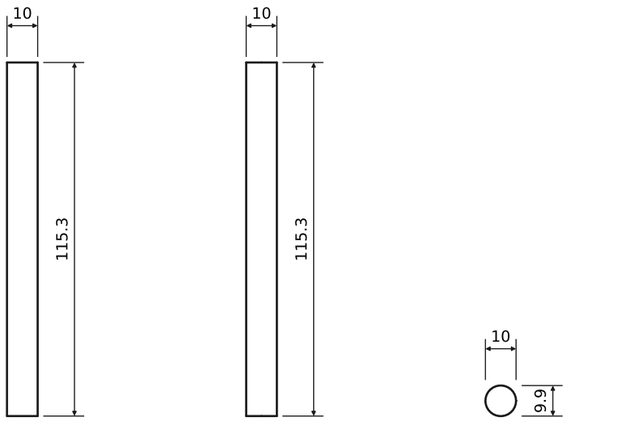
closes the pod behind the driver; a spigot locates it, three M4 screws hold it; the cable leaves through its centre



FRONT **FROM THE RIGHT** **FROM ABOVE**

14. rod-1, rod-2: lathe, stainless; 9 cm3 each; 4 for the set; 1:2

115 long; M8 thread 15 deep each end



FRONT **FROM THE RIGHT** **FROM ABOVE**

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

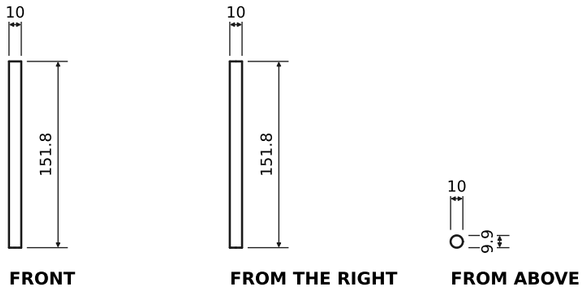
mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

15. rod-3: lathe, stainless; 12 cm3 each; 2 for the set; 1:5

152 long; M8 thread 15 deep each end



Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	18 mm Baltic birch plywood, BB/BB	420 x 880	26.80
2	top	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
3	bottom	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
4	front-panel	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 844	32.06
5	back	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 844	32.06
6	brace	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 360	13.68
7	woofer-ring	cut	2	6 mm MDF	436 x 436	4.28
8	port	print	2	ASA, FDM (outdoor, paintable)	89 cm3	14.22
9	woofer	buy	2	FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm)	393 frame, 356 cut-out, 165 deep	194.00
10	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm	315 x 120 x 55 plate, 291 x 96 recess	422.40
11	waveguide	cast	2	Tinted polyurethane casting resin, translucent	1513 cm3, 643 x 250 x 302	521.55
12	waveguide-cap	cast	2	Tinted polyurethane casting resin, translucent	117 cm3, 158 x 65 x 158	135.16
13	hf	buy	2	BMS 4550-16 compression driver, 1 in exit, polyester diaphragm, 16 ohm	123 across, 52 deep, 2 x M6 on 76.2	154.00
14	rod-1, rod-2	lathe	4	Stainless steel 304	20 x 20 x 125	16.50
15	rod-3	lathe	2	Stainless steel 304	20 x 20 x 162	26.94
16	bark	buy	4	Bark for laminating, a square metre: paper birch (Betula papyrifera) sheet from fallen or felled trees, or a smooth-barked gum's shed bark pressed flat (or a UV-printed veneer of it); approve a sample board first		[PRICE]
17	woofer-fixings	buy	2	M6 x 25 button heads (A2) with M6 T-nuts, a set of 8 (the 15PR400's eight holes on 375)		[PRICE]
18	hf-fixings	buy	2	M6 x 25 socket caps with washers (the compression driver to the horn's flange), a pair		[PRICE]
19	gasket	buy	4	1.5 mm closed-cell foam gasket strip, 1 m		5.00
20	damping	buy	4	Polyester fibre fill, 150 g		8.00
21	amp-fixings	buy	2	Hypex 4.3 x 16 self-tapping pan heads and 3 mm EPDM tape, a plate's set		10.00
22	cable	buy	6	Round-sheathed speaker cable 2 x 1.5 mm2, 1 m		2.00
23	terminals	buy	8	6.3 mm insulated push-on terminal		0.50
24	hf-pad	buy	2	Resistor L-pad for the HF channel, about -10 dB at 16 ohm (lowers the amplifier's hiss through a 113 dB driver)		[PRICE]

13 more lines: bom.md

Notes

1. Bark on the shell (sides, top, bottom and back): real birch bark sheet (Betula papyrifera, from fallen or felled trees) laminated to the plywood with PVA in a vacuum bag and sealed with a matt waterborne lacquer, wrapped round the shell's front edges; for eucalyptus, the shed bark of a smooth-barked gum (spotted or snow gum) pressed flat and laminated the same way, or a UV-printed veneer of it. Approve a sample board first.
2. Glue-up: sides to the bottom, the brace, then the front panel (set back 24 mm), the back, then the top.
3. The woofer flush in its rebate on a gasket, eight M6 T-nuts from inside, the ring over its frame; the port pushed in dry, trimmed to tune (impedance minimum at fb), then bonded.
4. The waveguide stands on three 10 mm stainless rods, M8 at both ends, in inserts in the top and in pads cast inside its underside.
5. Covers (options): a cloth grille, a slotted plywood grille (lenticels or strips), a cloth disc over the woofer alone, or the panel faced in bark or cork; a grille sits flush with the shell's front on four pegs into cups in the panel's corners, its back at least 12 clear of the cone at rest (twice the woofer's Xmax); the disc stands on the woofer's ring on magnets.

Parts list and notes

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-11 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15